

Movie Success Prediction and Analysis using Machine Learning

A data-driven approach to predicting movie revenue and uncovering the factors that drive box-office success — from budget and popularity to audience ratings.

Python & Libraries

Pandas, NumPy, Matplotlib, Scikit-learn — all in Google Colab.

Machine Learning

Linear Regression trained on movie features to predict revenue.

Data Analytics

Exploratory analysis and visualization of 5,000 TMDB movie records.

Movies Domain

Real-world box-office data sourced from the TMDB 5000 Kaggle dataset.

From Raw Data to Revenue Prediction

Dataset

Data Cleaning

Exploratory EDA

Linear Regression

The pipeline begins with the **TMDB 5000 Movie Dataset** from Kaggle. After cleaning and preprocessing — handling missing values, encoding genres, and normalizing features — we perform **exploratory data analysis** with visualizations to uncover patterns. The refined features then feed a **Linear Regression** model trained on an 80/20 train-test split to predict movie revenue.

01

Data Collection

TMDB 5000 Movie Dataset sourced from Kaggle with 4,800+ movie records.

02

Cleaning & Preprocessing

Handled nulls, encoded categorical fields, removed outliers.

03

Exploratory Data Analysis

Top-10 revenue films, budget vs. revenue, rating distribution charts.

04

Model Training

Linear Regression with train-test split for revenue prediction.

■ VISUALIZATIONS

Key Insights from Exploratory Analysis

Top 10 Highest Revenue

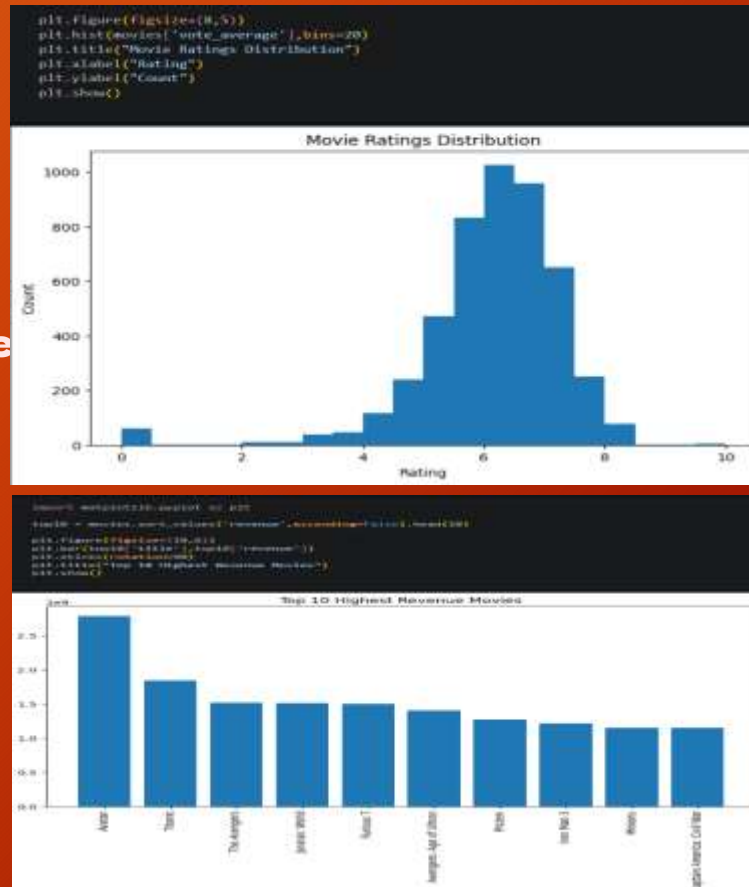
Blockbusters like *Avatar* and *Titanic* dominate the upper revenue tier, confirming the blockbuster phenomenon.

Rating Distribution

Most movies cluster between 6–8 average rating, showing a right-skewed bell curve with few extremes.

Budget vs. Revenue

A strong positive correlation — higher budgets generally yield higher gross returns, though variance increases at scale.



Conclusions & Future Scope

1

Prediction Model Built

Linear Regression successfully predicts movie revenue from budget, popularity, vote count, and rating features.

2

Key Drivers Identified

Budget and popularity show the strongest correlation with revenue, followed by audience ratings.

3

Skills Gained

Hands-on experience with data cleaning, Matplotlib visualization, and predictive analytics in Scikit-learn.

4

Future Enhancements

Random Forest / XGBoost models, sentiment analysis from reviews, and deployment as a web app.

THANK
YOU